



MECH MAINT- BOILER Department
SIMHADRI

Note No :1



Ref:SMTTP/MECH MAINT-/2023-24/REVIEW/967790

Date:

05/03/2024(12:21:07)

Subject:Review of LMI “INSPECTION OF BOILER TUBE WELD JOINTS BY PHASED ARRAY ULTRASONIC TESTING (PAUT)”

REF.: SMPP/O&M/MM/PP/

Date: 05.03.2024

Subject: Review of LMI “INSPECTION OF BOILER TUBE WELD JOINTS BY PHASED ARRAY ULTRASONIC TESTING (PAUT)”

LMI No: LMI/OGN/OPS/MECH/047

The above approved LMI was prepared in the month of Feb.’2019. The LMI was revised in the month of Dec ’2023. Changes in LMI are incorporated as per details at revision index of LMI are are marked in Bold .

Put up for approval please.

Submitted by:

राहुल जैन - उप महाप्रबंधक

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On:05/03/2024(12:21:07) **Ref:**SMTTP/MECH MAINT-/2023-24/REVIEW/967790

Note No:2

LMI with ref no- LMI/OGN/OPS/MECH/047 has been reviewed and the changes incorporated as per ref doc- "COS-ISO-00-OGN-OPS-MECH-047 Rev. No.: 02 Dec. 2023".

CA may please accord approval for this revised version of the subject LMI with ref no- "LMI/OGN/OPS/MECH/047".

Submitted by:

सुरेश बाबु कोल्ला - अपर महाप्रबंधक

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On:06/03/2024(10:34:30) **Ref:**SMTTPP/MECH MAINT-/2023-24/REVIEW/967790

Note No:3

Following revisions done in LMI as per revised OGN:

1. PAUT Sutiability elaborated at clause 7 (D).
2. For Mapping of PAUT of old weld joint, revised Annexure 1 incorporated

Submitted for approval of LMI (New LMI Doc) attached as annexure.

Submitted by:

ओम प्रकाश - अपर महाप्रबंधक

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On:06/03/2024(10:44:42) **Ref:**SMTTPP/MECH MAINT-/2023-24/REVIEW/967790

Note No:4

Pl examine.

Submitted by:

अर्जुन प्रदीप नायर - अपर महाप्रबंधक

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On:06/03/2024(15:47:50) **Ref:**SMTTPP/MECH MAINT-/2023-24/REVIEW/967790

Note No:5

As discussed pls review.

Submitted by:

वेंकटा श्रीनिवासा राव नुन्ना - उप महाप्रबंधक

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On:12/03/2024(12:58:57) **Ref:**SMTTPP/MECH MAINT-/2023-24/REVIEW/967790

Note No:6

May please review the points related to responsibility clause.

Submitted by:

सुरेश बाबु कोल्ला - अपर महाप्रबंधक

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On:12/03/2024(18:10:55) **Ref:**SMTTPP/MECH MAINT-/2023-24/REVIEW/967790

Note No:7

Necessary done.

Submitted please

Submitted by:

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On:13/03/2024(10:01:59) **Ref:**SMTTPP/MECH MAINT-/2023-24/REVIEW/967790

Note No:8

Submitted after making corrections as discussed. For further review please.

Submitted by:

सुरेश बाबु कोल्ला - अपर महाप्रबंधक

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On:13/03/2024(12:17:19) **Ref:**SMTTPP/MECH MAINT-/2023-24/REVIEW/967790

Note No:9

LMI reviewed. Forwarded for further approval pls.

Submitted by:

वेंकटा श्रीनिवासा राव नुन्ना - उप महाप्रबंधक

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On:13/03/2024(15:55:47) **Ref:**SMTTPP/MECH MAINT-/2023-24/REVIEW/967790

Note No:10

Submitted by:

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On:13/03/2024(18:04:38) **Ref:**SMTTPP/MECH MAINT-/2023-24/REVIEW/967790

Note No:11

Submitted by:

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On:13/03/2024(18:10:17) **Ref:**SMTTP/MECH MAINT-/2023-24/REVIEW/967790

Note No:12

Submitted by:

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On:13/03/2024(18:21:00) **Ref:**SMTTP/MECH MAINT-/2023-24/REVIEW/967790

Note No:13

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On:19/03/2024(15:58:04) **Ref:**SMTTP/MECH MAINT-/2023-24/REVIEW/967790

Note No:14

Submitted by:

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On:21/03/2024(06:43:10) **Ref:**SMTPP/MECH MAINT-/2023-24/REVIEW/967790

Note No:15

Approved & Submitted by:

संजय कुमार सिन्हा - मुख्य महाप्रबंधक

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On:22/03/2024(20:34:36) **Ref:**SMTPP/MECH MAINT-/2023-24/REVIEW/967790

NTPC
SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

LMI/OGN/OPS/MECH/047

ISSUE NO.1; REVISION NO. 2; DATE: December 2023

**INSPECTION OF BOILER TUBE WELD
JOINTS BY PHASED ARRAY
ULTRASONIC TESTING (PAUT)**

APPROVED FOR

***Implementation*-----**

HOP, SIMHADRI

Date-----

LMI APPROVAL DOCUMENTS

<u>1.0 TITLE OF LMI</u>	INSPECTION OF BOILER TUBE WELD JOINTS BY PHASED ARRAY ULTRASONIC TESTING (PAUT)
<u>2.a) DOCUMENT CODE</u>	<u>LMI/OGN/OPS/MECH/047/</u>
b) REVISION/ISSUE NO.&DATE	<u>ISSUE-1; REV: 2; DEC'2023</u>
<u>3.ORIGINAL/REF.DOCUMENT</u>	<u>COS-ISO-00-OGN/OPS/MECH/047/</u> <u>REV:0</u>
<u>4.PREPARED BY:</u> DGM (MM-PP)	SIGNATURE-----
<u>5. CHECKED BY :</u> AGM (MM-PP)	SIGNATURE-----
AGM (BMD)	SIGNATURE-----
GM (TS)	SIGNATURE-----
GM (O & M)	SIGNATURE-----
<u>6. REVIEWED AND APPROVED BY</u>	SIGNATURE:----- HOP, SIMHADRI
<u>7.DATE OF ISSUE</u>	
<u>8.INDEX LIST</u>	ENCLOSED.

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<u>2.0 INTRODUCTION</u> Weld joint failure is a significant factor contributing to Boiler tube leakages resulting in forced outages and availability loss. Typical boiler tubes have thickness range varying between 4 mm to 18 mm and diameter between 38 mm to 160 mm. Traditionally, radiography (RT) is the technique used for volumetric inspection of a weld joint's quality and integrity, but this technology has several disadvantages, such as (a) radiation hazards (b) need of work area isolation (c) less sensitive to linear defects (d) more time consuming (e) depth of discontinuity is not indicated (f) requires access to both sides of the components (g) A time gap between shot and evaluation result. Further, safety regulations are getting more and more severe worldwide, drastically restricting the use of radiography. Conventional Ultrasonic testing technique is being considered as an alternative to Radiography Testing for volumetric inspection of welds but use of this was also not feasible in boiler tubes due to lower thickness and size of the tubes. Phased Array Ultrasonic Testing (PAUT) is an advanced method of ultrasonic testing that has become popular in the NDT industry for its well-known advantages of easier, faster, and better accuracy along with permanent record of all collected data. With use of mechanical scanner along with specially designed ultrasonic phased array probes with suitably curved wedges, it is now possible for inspection of smaller diameter welds of lower thickness plain tubes (rifle tube excluded) 4.0 mm and above in boiler power plants. The objective of this LMI is to improve the reliability of boiler operation by the use PAUT technique for volumetric inspection of tube weld joints. <u>3.0 SCOPE</u> The scope of this LMI is to detail the methodology to be adopted while carrying out inspection of boiler tube weld joints by Phased Array Ultrasonic Testing (PAUT). <u>4.0 REFERENCE DOCUMENT</u> Operation guidance note : COS-ISO-00-OGN/OPS/MECH/047 Rev.0 circulated by operation services - NTPC Corporate center. <u>5.0 SUPERSEDED DOCUMENT</u> Operation Guidance Note- Inspection of Boiler Tube Weld Joints by Phased Array Ultrasonic Testing (PAUT): COS-ISO-00-OGN/OPS/MECH/047 Rev. No.:1, issued in September 2021		
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6.0 SPECIFIC INSTRUCTIONS

The following approach is to be followed to attain the objectives of this LMI:

During Overhauls, PAUT may be carried out for old weld joint inspection of boiler tubes covering different locations in Water wall, Superheater, Reheater, Economizer etc., in phased manner. PAUT may be continued in 100% old and 100% new joints with additional requirement of min 5% weld joint radiograph in new joints as per IBR.

b) If PAUT is planned for new weld joints, prior approval may be taken from Statutory authorities (CIB/ Director of Boilers/competent authority of the state).

7.0 METHODOLOGY OF IMPLEMENTATION AND RESPONSIBILITY CENTERS**A) PHYSICAL CONDITIONS REQUIRED FOR TESTING**

The minimum gap between tubes required (at least 25 mm) for movement of the scanner on tube periphery is to be ensured. If required, removal of coil binder strip/connector disengagement and pulling by inflatable jack may be carried out to create sufficient gap. Manual scanning with probe (not mounted on the scanner) can be done on part weld, in case of special needs like space constraints, fin welding and where only one side approach on the tube is available. For surface preparation, in case of vertical/horizontal coil joints, cleaning with emery paper is sufficient and in water wall/steam cooled wall tube, grinding/buffing and removal of any welding splatter is to be done.

Responsibility - Mech. Maint. - Pr. Parts group

B) PAUT EQUIPMENT AND CALIBRATION**Equipment, Probes & Wedges, Scanner, Encoders & Calibration Blocks**

Equipment : Phased Array Equipment with minimum 16:64 configuration

Probes & Wedges: Two numbers low profile suitable PAUT probes with optimized elevation having 7.5-10 MHz frequency, 16-32 element to use during testing. Low profile curved wedges set (to cover the entire range of OD to be tested and compatible to above probes) having irrigation holes and providing refraction angle of 50 to 60 degrees for shear wave in steel to be used. 7.5 MHz probe can be used for inspection of SS tubes up to 6.5 mm thickness. Dual linear array (dual 16 element array) of 5 MHz in transmitter-receiver longitudinal (TRL) configuration with compatible curved wedges can be used for inspection of SS tubes of thickness from 4 to 12 mm (OD in the range of 1 to 4 inch).

Scanner The two probes are to be mounted on the scanner capable to properly encircle.

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the tubes to be tested for inspection of the welds from both sides simultaneously. The scanner should have provision for coverage of the entire range of OD of boiler tubes to

be tested. Scanner will not be needed in case probe is to be moved manually by hand (due to any space constraint for movement of the scanner, like for water wall tubes having fin welding).

Position Encoder Use of the position encoder is a must for inspection with scanner. The encoder movement should be integrated to the movement of the probe.

Calibration /Reference Block For calibration of the speed and wedge delay, IIW V1/V2 block could be used. For TCG calibration, a cubical steel block of dimension 125x30x20 (in mm) could be used with two numbers of 1 mm dia. side drill holes. The holes should start at opposite faces at 4 mm & 8 mm depth and about one inch from respective edges extending up to 50% width. For the sensitivity calibration, tubes of similar material with tolerance of 10% in OD/thickness can be used. This tube of about 200 mm length should have 10% & 20% circumferential notches on ID & OD both, ~1.5 mm wide and one inch in length. 10% or 20% ID & OD notches can be on same circumference, located oppositely. The notches should be placed about 2-inch distance from the edges, so that there is ~100 mm axial distance between the circumferential notches.

Software Apart from the software for assisting in equipment set-up and calibration, following software also to be used: Software for weld-lay-out to be superimposed on scan signal and Software for data storage and analysis.

Calibration/System set-up the system comprising equipment, probe, wedge & encoder must be calibrated for performing the inspection. Scan plan, elemental check, speed, wedge delay, TCG, Sensitivity calibration, signal to noise ratio and calibration of the encoder to check the accuracy of the position . The calibration must be checked at least once prior to the examination everyday.

Responsibility - FQA & Mech. Maint. - Pr. Parts group

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TITLE: INSPECTION OF BOILER TUBE WELD JOINTS BY PHASED ARRAY ULTRASONIC TESTING (PAUT) REV: 2 DATE: DEC'23

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C) OPERATOR QUALIFICATION

It is to be ensured that the personnel carrying out PAUT are experienced in ultrasonic technique with Level II/Level III certification. The test personnel should also have suitable training in PAUT (preferably Level-2) for operation of the equipment and

interpretation of the PAUT signal.

Responsibility - FQA & Mech. Maint. - Pr. Parts group

D) NATURE OF DEFECTS & ACCEPTANCE CRITERION

PAUT is best suitable for any type of linear defect such as lack of penetration, lack of side wall fusion & cracks. It can also detect large size volumetric indication like blow hole/cluster porosity/slag inclusion etc. However, it is difficult to detect smaller size single porosity/pin hole porosity/scattered slag by PAUT. BTL weld joint failure analysis shows that liner defects and cluster of porosity in weld joints are the main reason of failures in units. Defect acceptance criterion for PAUT weld joint inspection would be same as for radiography weld joint inspection. Indications characterized as cracks, lack of fusion, or incomplete penetration are unacceptable regardless of length. If the indications exceed the reference level amplitude and have lengths which exceed 1/4 in. (6 mm) for t up to 3/4 in. (19 mm); 1/3t for t from 3/4 in. to 2-1/4 in. (19 mm to 57 mm); 3/4 in. (19 mm) for t over 2 1/4 in. (57 mm) Where, t is the thickness of the weld excluding any allowable reinforcement. For a butt weld joining two members having different thicknesses at the weld, it is the thinner of these two thicknesses. If a full penetration weld includes a fillet weld, the thickness of the throat of the fillet shall be included in it. Responsibility – Mech. Maint. - Pr. Parts group

PAUT Suitability

PAUT is used for inspection of Horizontal panels including Eco coils, LTSH coils, Re heater control stage), Vertical panels (Platen/Final SH/Radiant/Reheater tube) & Penthouse joints, coil fabrication work outside boiler areas and Water Wall panel plain tubes (to be inspected without cutting fins). Water wall panel tube weld joint to be inspected from both inside and outside in areas where weld joint failure occurred in the past.

Responsibility – Mech. Maint. - Pr. Parts group

E) CODES OF ACCEPTANCE

With the proven potential capabilities of phased array ultrasonic testing through various code cases, this technology has been brought into ASME Boiler & Pressure Vessel Code Section V, Article 4, 2010 Edition (2) for welds. ASME, Article 4, Mandatory Appendix V (2) calls for PAUT and the requirements are mentioned. Mandatory Appendix V, paragraph V-471 qualifies phased array sectorial scans and E-scans for encoded weld evaluations. ASME Boiler & Pressure Vessel Code, Section V, Article 23, SE-2700 provides guidelines for interpretation of different type of defects detected in PAUT.

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F) SCOPE OF WORK

2% Cross check of PAUT with RT is to be kept in SOW and in case of failure in identification of liner defect, root mismatch or cluster porosity by PAUT but detected by RT, all PAUT need to be repeated.

8.0 RESPONSIBILITIES

The responsibility of carrying out different tasks in relation to Inspection of Boiler Tube Weld Joints by PAUT as detailed in the LMI are with HOD (BMD) & HOD (FQA)

9.0 RECORD KEEPING

The records to be maintained under this LMI include.

- a) Compliance record of the activities detailed under this LMI are to be maintained as **per Annexure-I & II..**
- b) The final report consisting of test results and PAUT Inspection data

The above records are to be maintained by HOD (BMD).

10.0 REVIEW

This document shall be reviewed by Head of The Project on two yearly basis.

NTPC		LMI		SIMHADRI	
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<u>11.0 REVISION INDEX</u>					
Sl. No.	Rev. no.	Date	Details	Approved by	
1.	001	21.09.2021	Revision done as per COS- ISO-00- OGN/OPS/MECH/047 Rev. No.: 1, issued in September 2021		

NTPC	13.0 COMPLIANCE RECORD	SIMHADRI
TITLE: INSPECTION OF BOILER TUBE WELD JOINTS BY PHASED ARRAY ULTRASONIC TESTING (PAUT) REV: 2 DATE: DEC'23		
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UNIT NO.	ACTIVITY	COMPLIANCE

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SN	HEADER	DESCRIPTION	CHECKLIST
1	NAME OF AGENCY	M/S XYZ	Agency name filled: (Y/N)
2	PURPOSE OF TEST	Encoded phased Array inspection of weld in different areas of boiler (Eco, LTSH, PSH, Reheater, Water wall etc.) tube weld joints, data acquisition and subsequent analysis of data involving classification into planer and non planer defects.	WW: (Y/N) LTSH: (Y/N) ECO:Y/N) REHEATER: (Y/N) PSH: (Y/N) OTHER: Please specify(Y/N)
3	DOCUMENT REFERENCE NO	PAUT/STATION NAME/UNIT/OH/TUBE/WJ/April-21/29.04.2021	Document Reference Filled: (Y/N) No
4	COMPONENT IDENTIFICATION NO AND DESCRIPTION	AREA (ECO/LTSH/WW etc.)/TUBE NO OR COIL NO	Component identification no and Description filled: (Y/N)
5	AREA TO BETESTED	Full Weld Volume, HAZ and an appropriate amount of adjacent parent material.	(Y/N)

6	TEST SAMPLE (Y/N):	Should be available at site to vendor to calibrate and demonstrate capability of man and machine identify the defects in presence of NTPC PAUT qualified personnel (BMD/FQA).	(Y/N)
7	PERSONNEL (MINIMUM QUALIFICATION)	All personnel carrying out this written instruction should have a minimum ASNT/CSWIP/PCN Level 2 PAUT (Welds)	(Y/N)
8	SAFETY REQUIREMENTS	All operators should be familiar with plant safety requirements. Also, all the PPEs required for the job (as instructed by EIC) should be worn while carrying out the job.	(Y/N)
9	EQUIPMENT REQUIRED	Equipment model, Probe, Wedge, Calibration block, Encoder, Software	Equipment details filled (Y/N)

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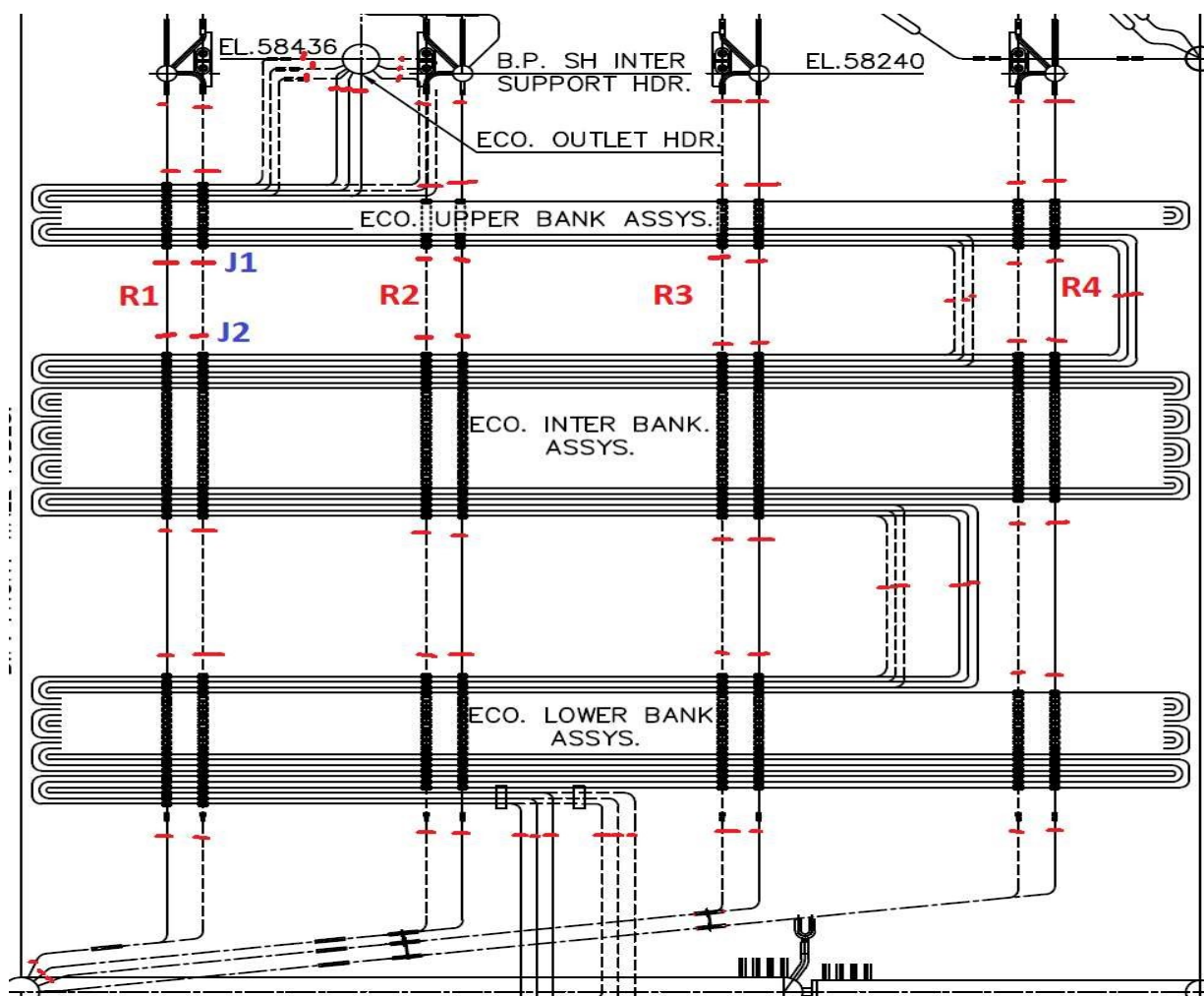
10	EQUIPMENT SETTINGS	Details regarding thickness, aperture, first element, minimum angle, max angle, angle step, focus depth, index offset, skew. Scan start, scan stop, scan resolution, Geometry, weld type, SW initial velocity, Scan offset, Acceptance criteria etc.	Equipment settings (Y/N) filled
11	PRE-TEST ACTIVITIES	Details regarding element check, velocity, sensitivity, wedge delay calibration and TCG calibration	Element Check:(Y/N) Velocity, Sensitivity, TCG Calibration: (Y/N)
12	DETAILED INSTRUCTIONS FOR APPLICATION OF TEST	Detailed process regarding process of test step wise	Step detailed (Y/N) wise filled:
13	POST TEST PROCEDURE S	Posttest activities like cleaning. Removal of markings etc.	Post activities(Y/N)

14	RECORDS	Data analysis must be carried out by Level 2 (PAUT) or higher. Sizing and nature of defect to be done and acceptance and rejection criteria to be done.	Data analysis by PAUT Level 2 done: (Y/N) Sizing, nature of defect and rejection criteria defined: (Y/N)
15	REPORTS	Report shall be made on a standard report form and consists of detailing of defects or "No defects found", Operator name, signature and certification, Component Sr no., Date.	Report in Standard format consisting of relevant details prepared: (Y/N)

INITIATOR SIGNATURE:

AUTHORISING PERSON SIGNATURE

ANNEXURE-II Mapping of Unit-X PAUT of old weld joint of Economiser/RH/SH/WW



SN	Overhauling duration	Area Covered	Coil/Hanger No.	Row No.	Joint No.	Remarks
1	FY 2022-23	Eco UB top	All coils/ xx to xx	x	J1/J2	All joints inspected/ Coil no. xxx to xxx inspected.
2	FY 2023-24					
3	FY 2024-25					
4						
5						